

A Comprehensive Review of the Chemical Composition and Toxicological Impact of Larsen Embalming Fluid in Medical Anatomy Laboratories: Implications for Neonatal Rat Surgery Preparation

Martin

Ria Margiana

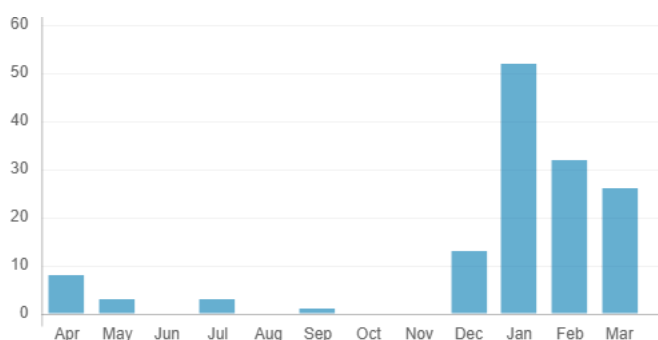
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Abstract

This systematic review aims to elucidate the chemical makeup of Larsen embalming fluid and its toxicological effects in anatomy laboratories, specifically with its use in newborn rat surgical preparation. The prevalent use of formaldehyde-based solutions for cadaver preservation effectively inhibits decay but presents considerable health risks, including respiratory inflammation, tissue discoloration, and potential carcinogenicity. The revised Larsen solution diminishes formaldehyde emissions by integrating chloral hydrate, glycerol, and certain inorganic salts, which assist in preserving natural tissue coloration, ensuring elasticity, and enhancing stability while reducing chemical hazards. These features are especially pertinent in the preparation of neonatal rat specimens for surgical training and anatomical research, where tissue integrity and minimized toxic exposure are essential. A thorough literature analysis was performed utilizing peer-reviewed academic databases such as PubMed, Scopus, the Web of Science, Google Scholar, and ScienceDirect to examine intricate chemical compositions and toxicological evaluations. This review emphasizes the efficacy and safety of this embalming formulation, providing insights into its prospective advantages for newborn surgical research, occupational health, and environmental preservation.

Downloads



References

- Bilge O, Celik S. Cadaver embalming fluid for surgical training courses: modified Larssen solution. *Surgical and Radiologic Anatomy*. 2017 Nov; 39:1263-72. <https://link.springer.com/article/10.1007/s00276-017-1865-4>
- Pekedis M, Yoruk MD, Binboga E, Yildiz H, Bilge O, Celik S. Characterization of the mechanical properties of human parietal bones preserved in modified Larssen solution, formalin and fresh frozen. *Surgical and Radiologic Anatomy*. 2021 Dec;43(12):1933-43. <https://link.springer.com/article/10.1007/s00276-021-02762-1>
- Khoshakhlagh AH, Mohammadzadeh M, Manafi SS, Yousefian F, Gruszecka-Kosowska A. Inhalational exposure to formaldehyde, carcinogenic, and non-carcinogenic risk assessment: A systematic review. *Environmental Pollution*. 2023 Aug 15;331:121854. <https://www.sciencedirect.com/science/article/pii/S0269749123008564>
- Kurniawan SB, Pambudi DS, Ahmad MM, Alfanda BD, Imron MF, Abdullah SR. Ecological impacts of ballast water loading and discharge: insight into the toxicity and accumulation of disinfection by-products. *Heliyon*. 2022 Mar 1;8(3). [https://www.cell.com/heliyon/fulltext/S2405-8440\(22\)00395-4](https://www.cell.com/heliyon/fulltext/S2405-8440(22)00395-4)
- Buser MC, Pohl HR, Abadin HG. Windows of sensitivity to toxic chemicals in the development of the endocrine system: an analysis of ATSDR's toxicological profile database. *International journal of environmental health research*. 2022 Feb 1;32(2):437-54.

<https://www.tandfonline.com/doi/abs/10.1080/09603123.2020.1772204>

Tesfaye S, Hamba N, Gerbi A, Neger Z. Oxidative stress and carcinogenic effect of formaldehyde exposure: systematic review & analysis. *Endocrinol Metab Syndr*. 2020 Sep 30;9(6):319-0. https://www.researchgate.net/profile/Solomon-T-Desissa/publication/351483377_Endocrinology_Metabolic_Syndrome_Oxidative_Stress_and_Carcinogenic_Effect_of_Formaldehyde_Exposure_Systematic_Review_Metabolic-Syndrome-Oxidative-Stress-and-Carcinogenic-Effect-of-Formaldehyde-Exposure-Systematic-Review-Analysis.pdf

Leider JP, DeBruin D, Reynolds N, Koch A, Seaberg J. Ethical guidance for disaster response, specifically around crisis standards of care: a systematic review. *American journal of public health*. 2017 Sep;107(9):e1-9. <https://ajph.aphapublications.org/doi/abs/10.2105/AJPH.2017.303882>

González-García A, Pinto-Carral A, Marqués-Sánchez P, Liebana-Presa C, García-Fernández R, Pérez-González S. Characteristics of the Competency Ethical Principles for the Nurse Manager: A Systematic Review. *Journal of Nursing Management*. 2025;2025(1):2575609. <https://onlinelibrary.wiley.com/doi/abs/10.1155/jonm/2575609>

Piasecki J, Walkiewicz-Żarek E, Figas-Skrzypulec J, Kordecka A, Dranseika V. Ethical issues in biomedical research using electronic health records: a systematic review. *Medicine, Health Care, and Philosophy*. 2021 Dec;24(4):633-58. <https://link.springer.com/article/10.1007/s11019-021-10031-6>

Drolet MJ, Rose-Derouin E, Leblanc JC, Ruest M, Williams-Jones B. Ethical issues in research: Perceptions of researchers, research ethics board members and research ethics experts. *Journal of Academic Ethics*. 2023 Jun;21(2):269-92. <https://link.springer.com/article/10.1007/s10805-022-09455-3>

Dal-Ré R. Waivers of informed consent in research with competent participants and the Declaration of Helsinki. *European journal of clinical pharmacology*. 2023 Apr;79(4):575-8. <https://link.springer.com/article/10.1007/s00228-023-03472-w>

Sangeetha VP, Arun V, Mohanan PV. Genotoxicity evaluation of nanosized materials. In *Biomedical Applications and Toxicity of Nanomaterials* 2023 May 9 (pp. 477-534). Singapore: Springer Nature Singapore. https://link.springer.com/chapter/10.1007/978-981-19-7834-0_19

Protano C, Buomprisco G, Cammalleri V, Pocino RN, Marotta D, Simonazzi S, Cardoni F, Petyx M, Iavicoli S, Vitali M. The carcinogenic effects of formaldehyde occupational exposure: a systematic review. *Cancers*. 2021 Dec 29;14(1):165. <https://www.mdpi.com/2072-6694/14/1/165>

Bobillot M, Delannoy V, Trouillard A, Kinowski JM, Sanchez-Ballester NM, Soulaïrol I. Potentially harmful excipients: state of the art for oral liquid forms used in Neonatology and Pediatrics Units. *Pharmaceutics*. 2024 Jan 17;16(1):119. <https://www.mdpi.com/1999-4923/16/1/119>

Honarvar Nazari, M., Mousavi, S.Z., Potapova, A., McIntyre, J. and Shi, X., 2021. Toxicological impacts of roadway deicers on aquatic resources and human health: A review. *Water Environment Research*, 93(10), pp.1855-1881. <https://onlinelibrary.wiley.com/doi/abs/10.1002/wer.1581>

Lopes J, Leuraud K, Klovov D, Durand C, Bernier MO, Baudin C. Risk of developing non-cancerous central nervous system diseases due to ionizing radiation exposure during adulthood: systematic review and meta-analyses. *Brain Sciences*. 2022 Jul 26;12(8):984. <https://www.mdpi.com/2076-3425/12/8/984>

Agrawal S, Volaity SS, Kilambi S, Kumar A, Neithalath N. A low-carbon approach for lime production using self-propagating high-temperature synthesis-driven limestone calcination. *Renewable and Sustainable Energy Reviews*. 2025 Mar 1;210:115192. <https://www.sciencedirect.com/science/article/pii/S1364032124009183>

Stabile AM, Pistilli A, Mariangela R, Rende M, Bartolini D, Di Sante G. New Challenges for Anatomists in the Era of Omics. *Diagnostics*. 2023 Sep 15;13(18):2963. <https://www.mdpi.com/2075-4418/13/18/2963>

Endale H, Mathewos M, Abdeta D. Potential causes of spread of antimicrobial resistance and preventive measures in one health perspective-a review. *Infection and drug resistance*. 2023 Dec 31;7:1515-45. <https://www.tandfonline.com/doi/abs/10.2147/IDR.S428837>

Austin HK, Schoenberg E, AUSTIN HK. A comprehensive literature review on the effects of formaldehyde on the upper respiratory tract. *Cureus*. 2024 May 6;16(5). <https://www.cureus.com/articles/235409-a-comprehensive-literature-review-on-the-effects-of-formaldehyde-on-the-upper-respiratory-tract.pdf>

Chrobak J, Iłowska J, Chrobok A. Formaldehyde-free resins for the wood-based panel industry: alternatives to formaldehyde and novel hardeners. *Molecules*. 2022 Jul 29;27(15):4862. <https://www.mdpi.com/1420-3049/27/15/4862>

Munn Z, Stone JC, Aromataris E, Klugar M, Sears K, Leonardi-Bee J, Barker TH. Assessing the risk of bias of quantitative analytical studies: introducing the vision for critical appraisal within JBI systematic reviews. *JBI Evidence Synthesis*. 2023 Mar 1;21(3):467-71. https://journals.lww.com/jbisrir/fulltext/2023/03000/assessing_the_risk_of_bias_of_quantitative.2.aspx

Jun H, Yoon SH, Roh M, Kim SH, Lee J, Lee J, Kwon M, Leem J. Quality assessment and implications for further study of autonomy: case reports using the case report guidelines and the Joanna Briggs Institute critical appraisal checklist. *Journal of Acupuncture Research*. 2021;38(2):122-33. <https://koreascience.kr/article/JAKO202116553540906.page>

Eiris A, Ramírez L, Couceiro I, Fernández-Fidalgo J, París J, Nogueira X. MLS-SPH-ALE: A review of meshless-FV methods and a unifying formulation for particle discretizations. *Archives of Computational Methods in Engineering*. 2023 Nov;30(8):4959-81. <https://link.springer.com/article/10.1007/s11831-023-09965-2>

Singh S, Bhardwaj S, Tiwari P, Dev K, Ghosh K, Maji PK. Recent advances in cellulose nanocrystals-based sensors: a review. *Materials Advances*. 2024;5(7):2622-54. <https://pubs.rsc.org/en/content/articlehtml/2024/ma/d3ma00601h>

Kennedy RM. Sedation in the emergency department: a complex and multifactorial challenge. *Pediatric Sedation Outside of the Operating Room: A Multispecialty International Collaboration*. 2021:413-73. https://link.springer.com/chapter/10.1007/978-3-030-58406-1_22

Patel S, Reinhardt ME, Roth S, Nova V. TXA in Liposuction to Minimize Bleeding Systematic Review Search Strategy. <https://scholarshare.temple.edu/handle/20.500.12613/7255>

Vara-Pérez M, Rossi M, Van den Haute C, Maes H, Sassano ML, Venkataramani V, Michalke B, Romano E, Rillaerts K, Garg AD, Schepkens C. BNIP3 promotes HIF-1 α -driven melanoma growth by curbing intracellular iron homeostasis. *The EMBO journal*. 2021 May 17;40(10): e106214. <https://www.embopress.org/doi/abs/10.15252/emboj.2020106214>

Roth S, Oyeyemi AL, Stoltenberg M, Gehris J, Moore T, Matthew J, Baishya ML. Search Strategies for a Systematic Review of Community Violence Exposure, Physical Activity, and Mental Health. <https://scholarshare.temple.edu/handle/20.500.12613/6310>

Guo Y, Lundebye AK, Li N, Ergon A, Pang S, Jiang Y, Zhu W, Zhao Y, Li X, Yao L, Wang L. Comparative assessment of food safety regulations and standards for arsenic, cadmium, lead, mercury and iodine in macroalgae used as food and feed in China and Europe. *Trends in Food Science & Technology*. 2023 Nov 1; 141:104204. <https://www.sciencedirect.com/science/article/pii/S0924224423003199>

Markam SS, Raj A, Kumar A, Khan ML. Microbial biosurfactants: green alternatives and sustainable solution for augmenting pesticide remediation and management of organic waste. *Current Research in Microbial Sciences*. 2024 Aug 13:100266.

Navitski I, Ramanaviciute A, Ramanavicius S, Pogorielov M, Ramanavicius A. MXene-based chemo-sensors and other sensing devices. *Nanomaterials*. 2024 Feb 28;14(5):447. <https://www.mdpi.com/2079-4991/14/5/447>

Okoroanyanwu U. Lithographic resists as amazing compact imaging systems–A review. *Micro and Nano Engineering*. 2024 Aug 14:100280. <https://www.sciencedirect.com/science/article/pii/S2590007224000431>

Rajput SK, Bhardwaj R, Madan S, Singh R. Ethylene Glycol: Industrial application and risk assessment. *Hazardous Chemicals*. 2025 Jan 1:205-17. <https://www.sciencedirect.com/science/article/pii/B9780323952354000281>

Basenach L, Renneberg B, Salbach H, Dreier M, Wölfling K. Systematic reviews and meta-analyses of treatment interventions for Internet use disorders: Critical analysis of the methodical quality according to the PRISMA guidelines. *Journal of behavioral addictions*. 2023 Mar 30;12(1):9-25. <https://akjournals.com/view/journals/2006/12/1/article-p9.xml>

Andreo-Martinez P, Ortiz-Martinez VM, Salar-Garcia MJ, Veiga-del-Bano JM, Chica A, Quesada-Medina J. Waste animal fats as feedstock for biodiesel production using non-catalytic supercritical alcohol transesterification: A perspective by the PRISMA methodology. *Energy for Sustainable Development*. 2022 Aug 1;69:150-63. <https://www.sciencedirect.com/science/article/pii/S0973082622000916>

Page MJ, Sterne JA, Higgins JP, Egger M. Investigating and dealing with publication bias and other reporting biases in meta-analyses of health research: A review. *Research synthesis methods*. 2021 Mar;12(2):248-59. <https://onlinelibrary.wiley.com/doi/abs/10.1002/jrsm.1468>

Gull M, Pasek MA. The role of glycerol and its derivatives in the biochemistry of living organisms, and their prebiotic origin and significance in the evolution of life. *Catalysts*. 2021 Jan 10;11(1):86. <https://www.mdpi.com/2073-4344/11/1/86>

Kandi V, Vadakedath S. Clinical trials and clinical research: a comprehensive review. *Cureus*. 2023 Feb 16;15(2). <https://www.cureus.com/articles/128436-clinical-trials-and-clinical-research-a-comprehensive-review.pdf>



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